

Andy Widjaja

Chicago, IL | andywidjaja.com | linkedin.com/in/andy-widjaja-aws | github.com/Sanctum-Origo-Systems/patina

Summary

Software engineer with 10+ years of experience building production systems, from sub-millisecond trading gateways processing 150K contracts/day to enterprise AI platforms handling 750M+ document processing. Actively building and testing Patina, an open-source cognitive agent framework on Claude Agent SDK with belief graph architecture, graduated autonomy, and MCP integration. At AWS, built and shipped a production agentic system (5,000+ global active users, 90%+ org adoption) and published an open-source MCP server (**16,000+ monthly installs**). Extensive experience in agent orchestration, distributed systems, API design, and evaluation infrastructure.

Technical Skills

Python, TypeScript, Java, C#, SQL | Agent orchestration, MCP protocol, LLM evaluation | Distributed systems, event-driven architecture, API design | PostgreSQL, OpenSearch, Neo4j, Redis | Docker, Kubernetes, serverless compute | CI/CD, infrastructure-as-code

Open Source

Patina: Cognitive Agent Framework (2026) | github.com/Sanctum-Origo-Systems/patina

Cognitive agent framework implementing belief graphs, graduated autonomy, and persistent memory via MCP with three-layer architecture: messaging gateway, Claude Agent SDK runtime, and cognitive MCP server. 25,000+ lines Python, 900+ tests.

- Created belief extraction pipeline: An LLM-based entity/relationship extraction from unstructured inbound messages into a queryable graph
- Applied a graduated autonomy model where agent decision-making authority grows or shrinks based on measured confidence per domain
- Implemented MCP server exposing 40+ tools for memory, prioritization, profiling, and introspection to maintain portability across different agentic clients

Experience

AWS, APEX Agentic AI Team | 2022–Present

Agent Infrastructure (2025–2026):

- Built and shipped Delivery Agent, a multi-step agent orchestration system for software delivery workflows (Python, TypeScript, MCP protocol, LLM reasoning). Scaled from 0 to 5,000+ global active users (90%+ org adoption) across 6 regions within 8 months and led LATAM and APJC technical roadshows (10 countries, 900+ engineers) to drive adoption from builders
- Co-authored ATLAS, an automated evaluation framework measuring agent output quality via precision/recall/F1. Deployed as CI/CD quality gate across 2,500+ concurrent users. Submitted to Amazon Science for peer review

- Published document-loader MCP server (awslabs, open source): structured content extraction from PDF/DOCX/XLSX/PPTX/images. [16,000+ monthly downloads](#)
- Developed Python dev tools: spec-package diff engine (LLM + structural comparison, 10 min vs. 4 hours manual), cross-framework workflow sync tool (1 sprint compressed to 1 day)
- Designed S&P Global agent migration architecture: LangGraph agents and MCP tools to AgentCore Runtime with state migration strategy

Enterprise AI Systems (2022–2025):

- Shipped AI-powered document intelligence platform for AIG: 750M+ documents, 165K daily ingestion, multi-region (NAMER/EMEA/APAC). OpenSearch + Textract pipeline with sensitive payment detection
- Developed production RAG system for DTCC (public-facing): Claude 3.5 Sonnet, cross-region inference, 60+ concurrent users, financial regulatory compliance
- Built authorization system for FIS Global using Amazon Verified Permissions: 30x cost reduction, multi-tenant RBAC/ABAC

Legal & General Investment Management America | 2018–2022

Full Stack Developer

- Built serverless event-driven portfolio performance calculation platform for quantitative trading team (Lambda, ECS, Kafka) and global account data management service serving all business units

Performance Trust Capital Partners | 2013–2018

Lead Developer

- Built proprietary bond analytics engines: price-yield analysis across CMOs, MBS, munis (Intex, Yield Book, Bloomberg APIs). Trained ML models for pricing and asset classification (Python, R)
- Decomposed legacy monolith into 20+ microservices. Redis, RabbitMQ, Docker orchestration

R.J. O'Brien & Associates | 2010–2013

Trading Systems Developer

- Built CME Globex trading gateway: 25,000 orders/day, 150,000 contracts, sub-second execution. FIX/FAST protocol
- Developed custom routing gateway for Morgan Stanley, order-type-based execution logic

Earlier Career | 2003–2010

Lead Developer at Newedge (futures clearing), Dell Professional Services, SEI Information Technology

Education & Speaking

BS Computer Science, University of Wisconsin-Madison

Speaking: re:Invent 2025 (Delivery Agent launch), re:Invent 2024 (breakout + chalk talk), re:Invent 2023, LATAM Agentic AI Tour 2026 (4 countries, 300+ engineers), APJC Agentic AI Tour 2026 (6 countries, 600+ engineers)